ALGEBRA 2 · UNIT 7: RATIONAL FUNCTIONS

Mural Business

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students start a mural-painting business. They model average cost with a rational function and interpret its asymptotes, solve work-rate and travel problems with rational equations, simplify rational expressions, use inverse variation, and set prices for their own business.

Average cost gives the horizontal asymptote meaning (the cost per mural can never go below $150), and work problems show why rates (not times) add.

Standards Addressed

- A-APR.6 Rewrite rational expressions
- A-REI.2 Solve rational equations and identify extraneous solutions
- F-IF.7d Graph rational functions with asymptotes
- A-CED.1 Create equations from context

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a mural photo. Part 1 average cost.
Day 2	Part 2 work and travel equations.
Day 3	Part 3 business plan.

Materials

- Student handout
- Graphing calculators or Desmos
- Optional: photos of local murals

How the Three Versions Differ

Level	What changes
CP2	Average cost table and asymptote meaning, work rates with unit fractions, and inverse variation.
CP1	Simplifying rational expressions, a travel rational equation, and a third-painter equation.
Honors	Rational inequalities, a work problem leading to a quadratic, and a design with break-even pricing.

Watch For (Common Misconceptions)

- Adding times instead of rates.
- Canceling terms instead of factors.
- Forgetting to exclude values that make a denominator 0.

Extension Ideas

- Get real price quotes from local artists.
- Graph profit per mural.
- Research public art funding.

Answer Key – CP2

Part 1: Average Cost

#	Task	Answer
1	Complete the table.	1,350; 390; 270; 190; 170
2	What value does C(x) get close to as x gets large? Why does that make sense?	$150, the material cost; the $1,200 is spread over more murals.
3	How many murals must you paint for the average cost to be $190?	1200 + 150x = 190x → x = 30 murals
4	Why is x = 0 not in the domain?	You can't divide by 0 (no murals means no average).

Part 2: Painters & Rates

#	Task	Answer
5	What fraction of a mural does Ben paint per hour?	⅑ of a mural per hour
6	Working together, how long do Ana and Ben take? Solve ⅛ + ⅑ = 1 ÷ t.	⅛ + ⅑ = 5 ÷ 18, so t = 18 ÷ 5 = 3.6 hours
7	With Cara, all three finish in 2.4 hours. How long would Cara take alone?	1 ÷ 2.4 − 5 ÷ 18 ≈ 0.1389 → Cara takes 7.2 hours
8	A big wall takes 24 painter-days: days = 24 ÷ (painters). How many days with 4 painters? With 6?	6 days; 4 days

Part 3: Business Plan

#	Task	Answer
9	Choose a price per mural. How many murals do you need so that the price is at least $50 more than the average cost?	Answers vary.
10	Choose a team and find how long one mural takes together.	Answers vary.
11	Write a short business plan.	Answers vary.

Answer Key – CP1

Part 1: Average Cost

#	Task	Answer
1	Rewrite C(x) as 150 + 1200 ÷ x. Identify the vertical and horizontal asymptotes and explain them in context.	Vertical asymptote x = 0 (no murals); horizontal asymptote y = 150 (the per-mural material cost).
2	How many murals give an average cost of $175?	1200 + 150x = 175x → x = 48 murals
3	A larger studio has $C_2(x) = (3000 + 120x) \div x$. For how many murals is the larger studio cheaper per mural?	$(3000 + 120x) \div x < (1200 + 150x) \div x \rightarrow 3000 + 120x < 1200 + 150x \rightarrow x > 60$ murals

Part 2: Rational Equations

#	Task	Answer
4	Ana takes 6 h, Ben 9 h. Together with Cara they take 2.4 h. Find Cara's time with a rational equation.	$\frac{1}{6} + \frac{1}{9} + 1 \div c = 1 \div 2.4 \rightarrow 1 \div c = 5 \div 12 - 5 \div 18 = 5 \div 36 \rightarrow c$ = 7.2 hours
5	You drive 60 miles to a job and back. You drive 10 mph faster coming home, and the round trip takes 2.7 hours of driving. Find your speed going.	$60 \div r + 60 \div (r + 10) = 2.7 \rightarrow 9r^2 - 310r - 2000 = 0 \rightarrow r = 40$ mph (reject the negative root)
6	Simplify $(x^2 - 9) \div (x^2 + 5x + 6)$. State the restrictions and where the graph has a hole.	$(x - 3)(x + 3) \div ((x + 2)(x + 3)) = (x - 3) \div (x + 2)$; $x \neq -2, -3$; hole at x = −3

Part 3: Business Plan

#	Task	Answer
7	Set a price and find the break-even number of murals with a rational equation.	Answers vary.
8	Schedule a three-person team and find their combined time.	Answers vary.
9	Write the plan with a graph.	Answers vary.

Answer Key – Honors

Part 1: Cost Analysis

#	Task	Answer
1	Solve $C(x) < 175$ and interpret the answer.	$x > 48$ (more than 48 murals)
2	Find where $C(x) = C_2(x)$ and explain which studio to choose for different seasons.	$1200 + 150x = 3000 + 120x \rightarrow x = 60$. Fewer than 60: small studio; more than 60: large studio.
3	Graph both and describe all asymptotes, intercepts, and end behavior.	C: VA $x = 0$, HA $y = 150$. C_2: VA $x = 0$, HA $y = 120$. No x-intercepts for $x > 0$; both decrease toward their asymptotes.

Part 2: Hard Equations

#	Task	Answer
4	One painter takes 5 hours longer than another. Together they finish in 6 hours. Find each painter's time.	$1 \div t + 1 \div (t + 5) = \frac{1}{6} \rightarrow t^2 - 7t - 30 = 0 \rightarrow t = 10$ (reject -3). Times 10 h and 15 h.
5	Solve $60 \div r + 60 \div (r + 10) = 2.7$ and interpret both roots.	$r = 40$ mph or $r = -50 \div 9 \approx -5.6$ (not a speed, so extraneous in context).
6	Solve $x \div (x - 3) - 3 \div (x + 3) = 18 \div (x^2 - 9)$.	Multiply by $(x - 3)(x + 3)$: $x(x + 3) - 3(x - 3) = 18 \rightarrow x^2 = 9 \rightarrow x = \pm 3$, but both make a denominator 0. No solution.

Part 3: Design a Pricing Model

#	Task	Answer
7	Find the profit per mural, $P(x) - C(x)$, and simplify into one rational expression.	Answers vary by simplification: $250 - 600 \div (x + 2) - 1200 \div x$
8	For what order sizes is the profit per mural at least \$100? (Use a table or solve.)	Test values: $x = 10$: $250 - 50 - 120 = 80$; $x = 15$: $250 - 35.3 - 80 \approx 134.7$; $x = 12$: $250 - 42.9 - 100 \approx 107.1$; $x = 11$: $250 - 46.2 - 109.1 \approx 94.7$. At least 12 murals.
9	Write your recommendation.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected. Goes beyond the minimum with a creative, realistic choice.	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Polished, organized, and visually engaging.	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.